

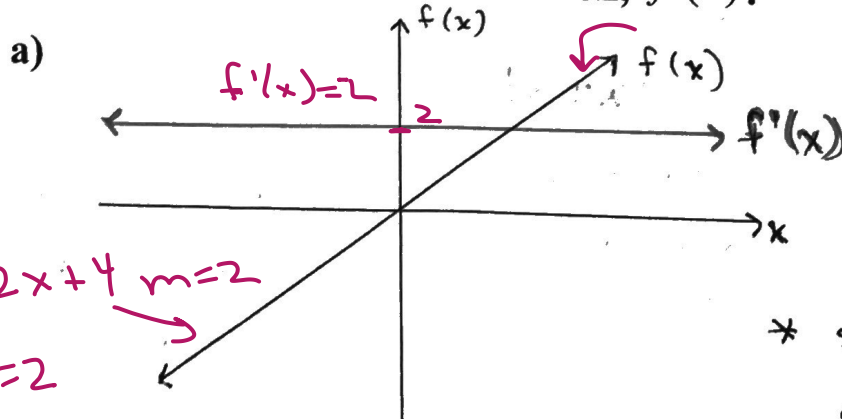
Graphing Derivatives

We can relate the graph of a function with its derivative. To do so, remember:

- i) The derivative of a function represents the slope of the tangent line of the function at any defined point, x .
- ii) For polynomial functions, each time we differentiate we reduce the degree of the polynomial by 1.

- Notes:
1. If the slope of the tangent is negative at point $x = a$ (ie: $f'(x) < 0$), then the derivative graph is below the x -axis.
 2. If the slope of the tangent is positive at point $x = a$ (ie: $f'(x) > 0$), then the derivative graph is above the x -axis.
 3. If the slope of the tangent is zero at point $x = a$ (ie: $f'(x) = 0$), then the derivative graph is on the x -axis.

Example 1: For each of the following graphs of $f(x)$ sketch the derivative function, $f'(x)$.

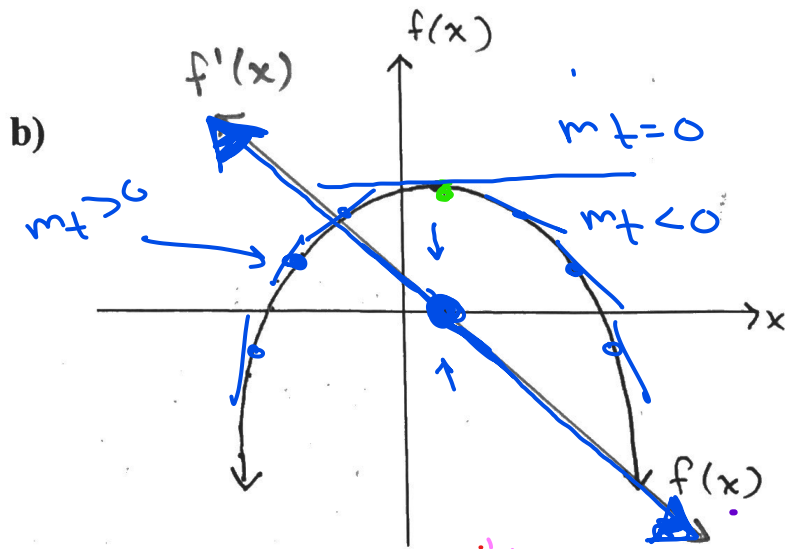


notes:

$f(x)$ is linear

$f'(x)$ is constant

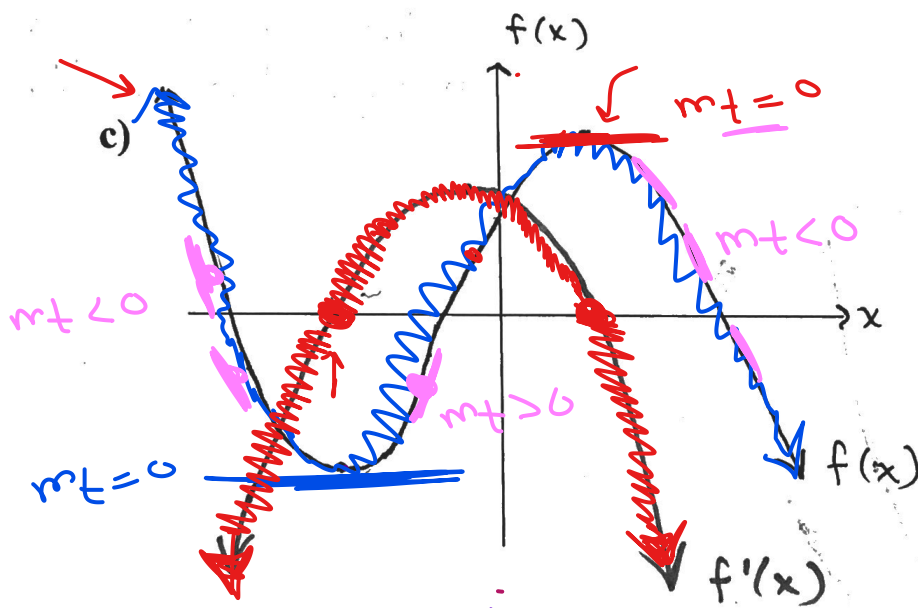
* slope is constant at every point.



notes:

$f(x)$ is quadratic

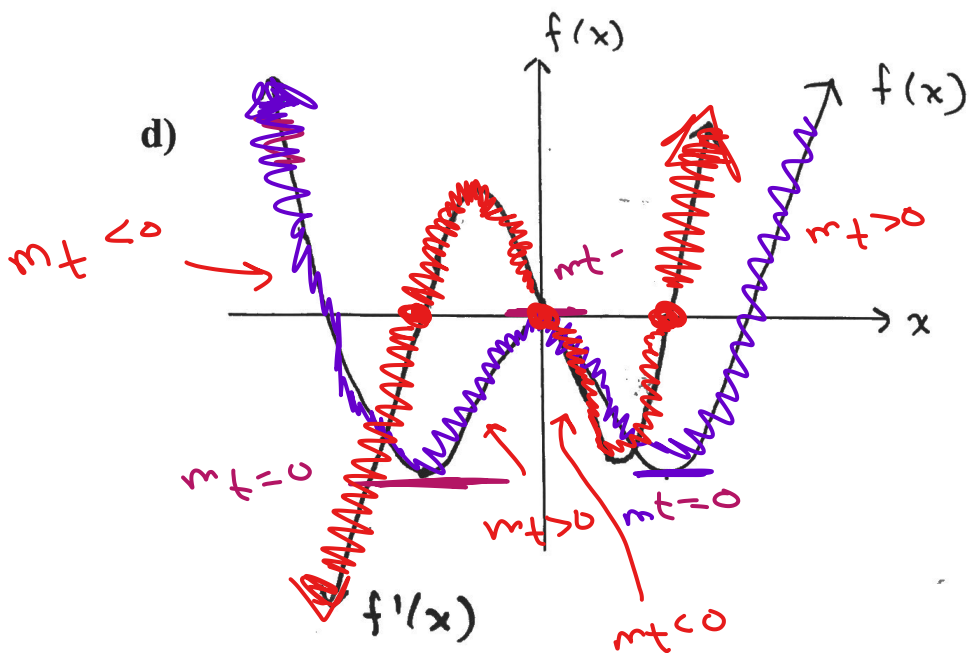
$f'(x)$ is linear



notes:

$f(x)$ is cubic

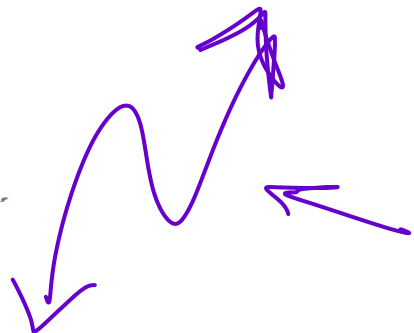
$f'(x)$ is quadratic

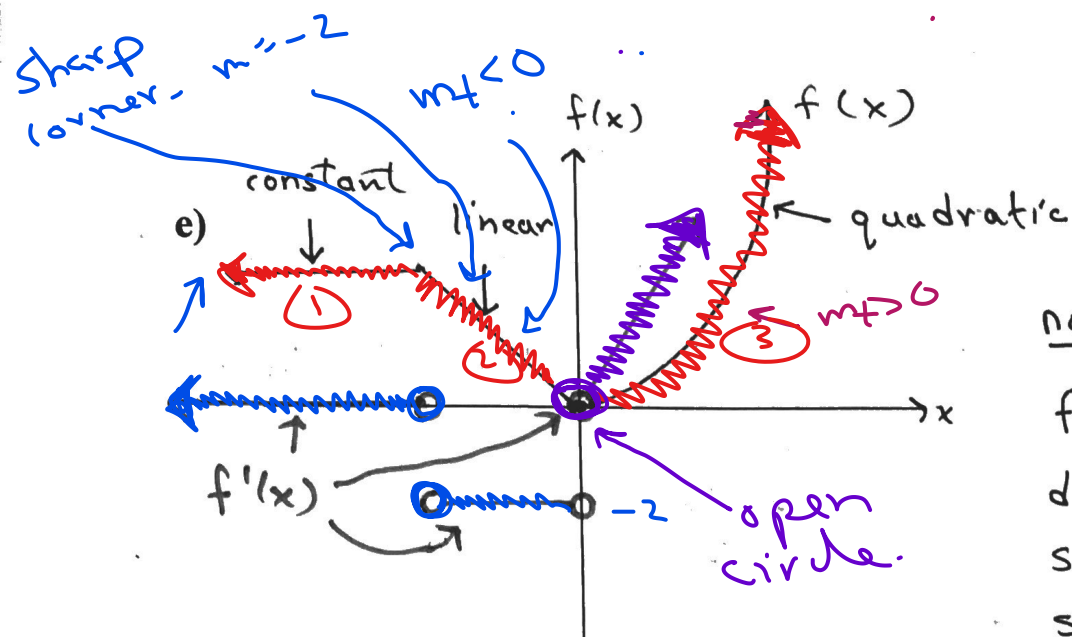


notes:

$f(x)$ is quartic

$f'(x)$ is cubic





notes:

$f'(x)$ is not defined at sharp corners since there is no unique tangent line.

Example 2: Given $f'(x)$ below, sketch a possible curve for $f(x)$.

